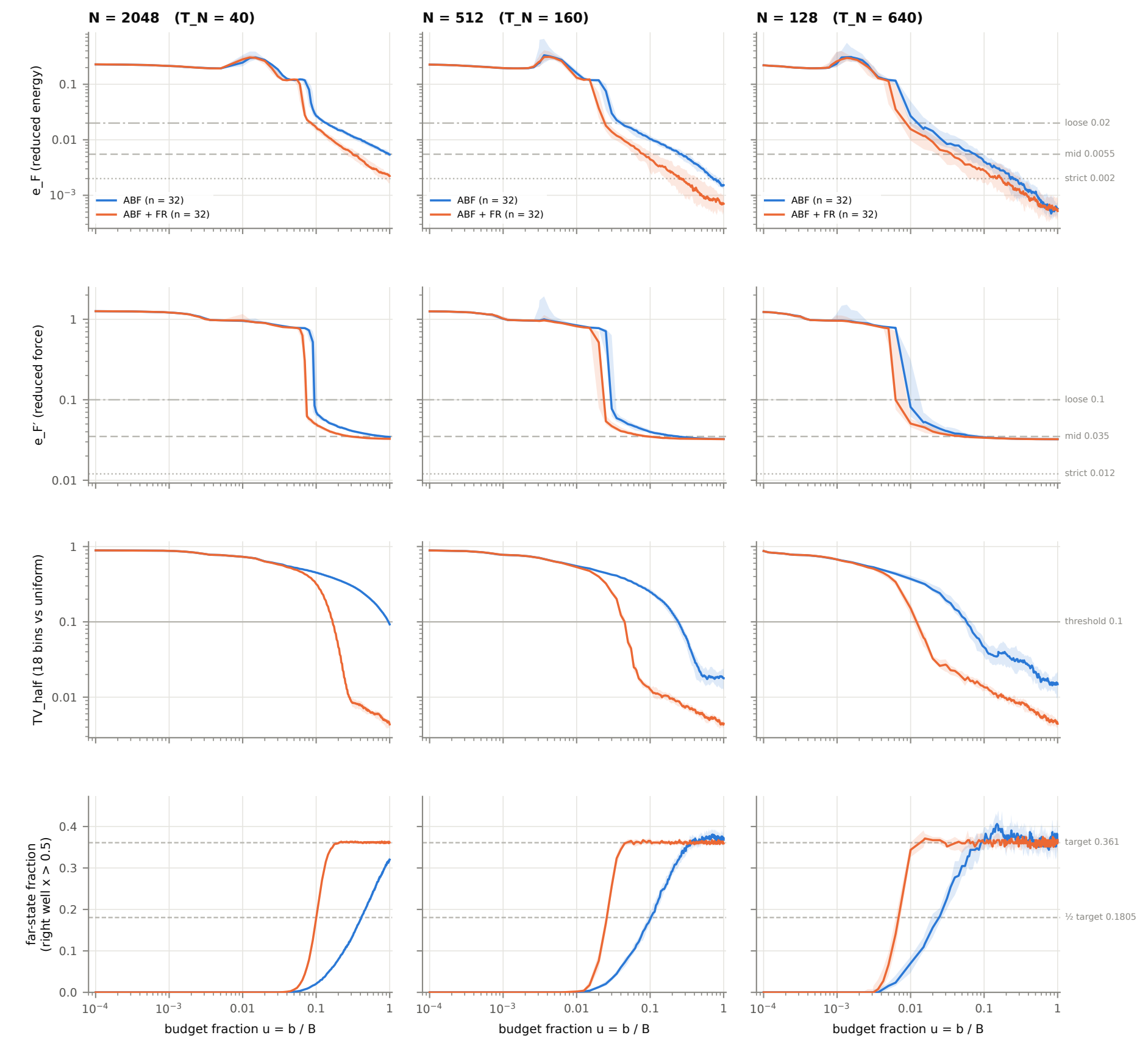


gateway alpha1: convergence curves on the normalised budget axis, selected N

Equal total budget per arm per seed:  $B = N \times n\_steps = 3.2768e+09$  walker-steps at every N ( $h = 2.5e-05$ ,  $T\_N = B \cdot h / N$ ); 32 seeds per (N, method). Common axes per row; every column spends the same B.



gateway alpha1 · S7\_convergence\_selected\_N · status: all planned (N, seed, method) complete · FR is not defined at N = 1 (ABF only, by design).  
Selected N: largest planned N; the middle of the sorted planned ladder (index  $\lfloor n/2 \rfloor$ ); the smallest N with an FR arm; N = 1 -- chosen from the PLAN, independent of the data (duplicates dropped).  
Lines: median over seeds at every save (24 log-spaced  $u < 5e-3$ , the 200 uniform  $u$ , physical checkpoints); bands: IQR. Grey lines (labelled at the right): frozen thresholds (dotted strict, dashed mid, dash-dot loose; TV\_half solid) and the far-state target / 1/2 target.  
Source results/mechanism/matched\_free\_energy/alpha1/analysis/summary.json (generated 2026-10-10T14:19:18Z, eqb\_metrics/2, 10000 bootstrap resamples); plot\_synthesis/2.